Metallacycles

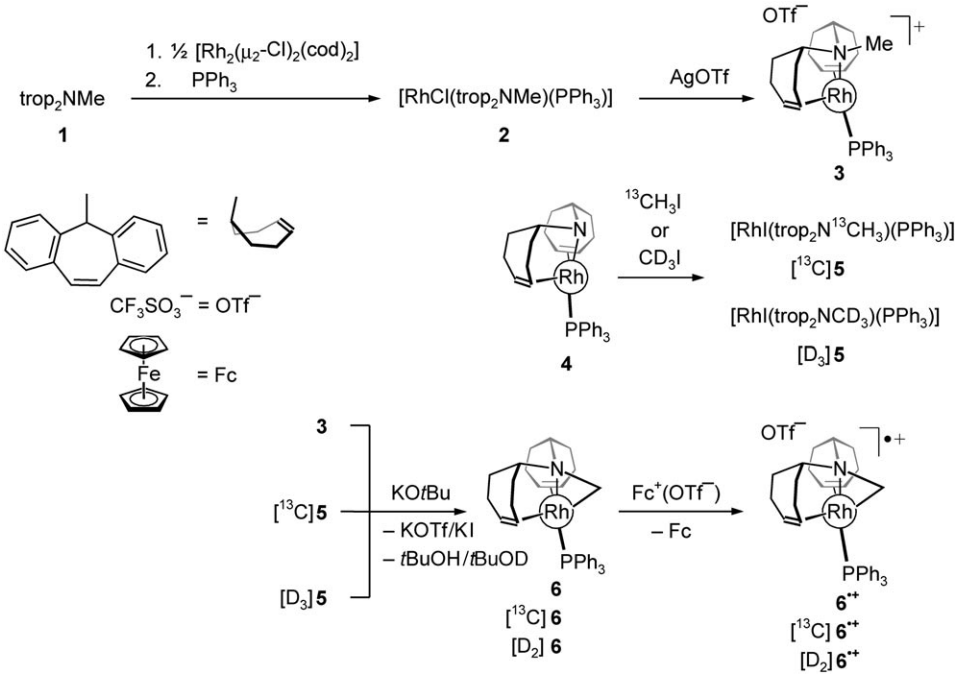
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Synthesis of a Rhodaazacyclopropane and Characterization of Its Radical Cation by EPR Spectroscopy**

Pascal Maire, Anandaram Sreekanth, Torsten Büttner, Jeffrey Harmer, Igor Gromov, Heinz Rüegger, Frank Breher, Arthur Schweiger, and Hansjörg Grützmacher**

Cyclopropane and its derivatives $(ER_2)_3$ (E = C, Si, Ge, Sn), bicyclobutanes, and other strained hydrocarbons have remarkably low oxidation potentials when compared to their unstrained linear counterparts.[1–3] Can this property be translated to transition-metal complexes, such that small rings incorporating the transition metal lead to more facile oxidation? With this question in mind, we characterized a d^8 Rh complex that has a metal center incorporated in a three-membered ring. Furthermore, we report the results from high-resolution pulse EPR spectroscopy, which, in combination with density functional theory (DFT) calculations, allowed us to characterize in detail the electronic structure of the corresponding radical cation.

In analogy to the preparation of $trop_2NH$[4] (trop = tropylidenyl = 5*H*-dibenzo[*a*,*d*]cyclohepten-5-yl), we prepared the *N*-methyl derivative $trop_2NMe$ (**1**) from $(Me_3Si)_2NMe$ and tropCl (Scheme 1). Subsequently, **1** was treated with $[Rh_2(\mu_2\text{-}Cl)_2(cod)_2]$ (cod = 1,5-cyclooctadiene) to yield $[Rh_2(\mu_2\text{-}Cl)_2(trop_2NMe)_2]$, which was treated with PPh_3 to give $[RhCl(trop_2NMe)(PPh_3)]$ (**2**). Addition of AgOTf ($OTf^- = CF_3SO_3^-$) led to the isolation of $[Rh(trop_2NMe)(PPh_3)]OTf$ (**3**) as red needles in 78% overall yield (for details, see the Supporting Information). The ^{13}C- and D_3-labeled iodide complexes [^{13}C]**5** and [D_3]**5** were prepared from the reactions of the rhodium(I)amide **4**[5] with $^{13}CH_3I$ and CD_3I, respectively. The NMR data indicate that **2** and **5** form trigonal bipyramidal 18-electron complexes in $CDCl_3$ solution with the halide atoms and the $C{=}C_{trop}$ units in the equatorial positions. Complex **3** forms an ion pair in solution, such that the OTf^- ion binds remotely from the metal in a pocket between the NMe group and the benzylic protons of the $trop_2NMe$ ligand.



Scheme 1. Synthesis of complexes **3**, [^{13}C]**5**, [D_3]**5**, **6**, [^{13}C]**6**, [D_2]**6**, and radical cations **6**$^{\bullet+}$, [^{13}C]**6**$^{\bullet+}$, and [D_2]**6**$^{\bullet+}$.

An ion pair is also observed for **3** in the solid state, again with the OTf^- ion bound remotely from the metal center and embedded between the arene groups of the tropylidenyl and PPh_3 ligands (Figure 1b).[6] The $[Rh(trop_2NMe)(PPh_3)]^+$ ion has an unusual "saw-horse" structure (Figure 1a) which is nonplanar and resembles a trigonal bipyramid with one equatorial ligand missing (selected bond lengths and angles are listed in Table 1). Structures of this type have rarely been observed for stable 16-electron complexes; some examples include the neutral amide **4**[5] and the Ru complexes $[Ru(CO)_2L_2]$ (**I**) and $[Ru(CO)(NO)L_2]^+$ (**II**; L = P*t*Bu_2Me).[7] The N-Rh-P bond angle in **3** (158.2(1)°) and the P-Ru-P bond angle in **II** (157.3°) are slightly more acute than the corresponding bond angles in the neutral complexes **4** (166.2°) and **I** (165.6°). The ct-Rh-ct angle (134.5(2)°; ct = centroid of the coordinated double bond) is similar to the corresponding angles in **4** and **I** (ca. 135°), but larger than that

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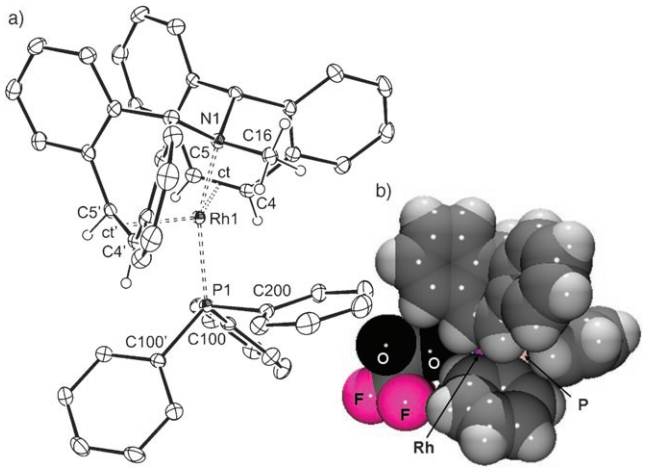


Figure 1. Structure of **3**·3 CH_2Cl_2. a) Thermal ellipsoids are set at 30% probability. The three CH_2Cl_2 molecules, the OTf^- counteranion, and the hydrogen atoms, apart from those in the *N*-methyl group and at the olefinic carbon atoms, are omitted for clarity. Selected bond lengths and angles are listed in Table 1 (ct = centroids of the C=C_{trop} units). Additional angles [°]: C100-P1-C100′ 102.4(3), C200-P1-C100/100′ 108.0(2), Rh-P1-C100/100′ 124.4(1). b) Space-filling model of **3** showing the position of the anion.

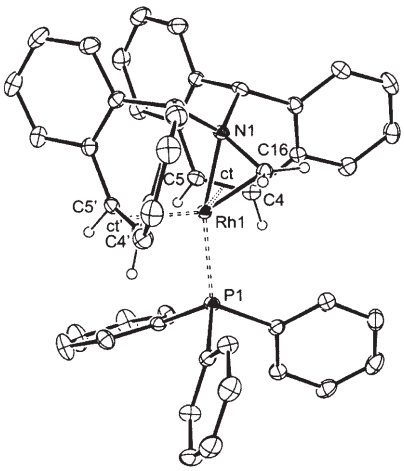


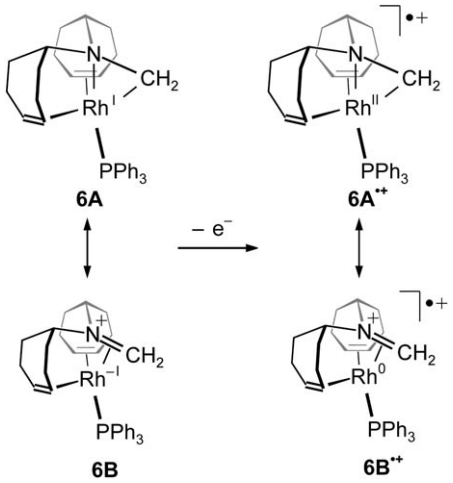
Figure 2. Structure of **6**·toluene. Thermal ellipsoids are set at 30% probability. The toluene molecule and hydrogen atoms, apart from those in the methylene bridge and at the olefinic carbon atoms, are omitted for clarity. Selected bond lengths and angles are listed in Table 1 (ct = centroids of the C=C_{trop} units).

in the nitrosyl complex **II** (120°). The small Rh1-P1-C200 angle (86.1(2)°) and relatively short Rh···C200 distance (2.80 vs. > 3.6 Å for the other Rh···C_{phenyl} distances) in **3** are especially noteworthy and indicate a hyperconjugative interaction between the P–C bond and the metal center.

The triflate complex **3** reacts cleanly with KO*t*Bu to give complex **6**. The compounds [^{13}C]**6** and [D_2]**6**, in which the *N*-methylene group is ^{13}C- and D-labeled, were obtained from the respective iodide complexes [^{13}C]**5** and [D_3]**5**. The result of an X-ray structure analysis on orange crystals of **6** is shown in Figure 2; selected bond lengths and angles are given in Table 1.

Interactions between a metal M and an unsaturated ligand X=Y can be considered with regard to the limiting cases represented by the heterocyclic metallacyclopropane form (type **A**, Scheme 2) and a X=Y→M donor–acceptor complex (type **B**).[8] Complex **6** can be described either as a rhodaazacyclopropane **6A** or a rhodium iminium ion complex **6B**. In the first case, the Rh center has a formal oxidation state of +1, and in the latter case a formal oxidation state of −1.

The Rh–N bond in **6** (2.059(3) Å) is shorter than in **3** (2.131(5) Å). Also, the N–CH_2 bond in **6** (1.446(5) Å) is significantly shorter than the N–CH_3 bond in **3** (1.505(7) Å), but considerably longer than the N=C bonds in iminium ions (1.274–1.301 Å)[9] and close in length to the N–C bonds in aziridines (1.440–1.471 Å).[10] Hence, the structural data favor



Scheme 2. Selected resonance structures for **6** and **6**$^{\bullet+}$.

the rhodaazacyclopropane description **6A**, which is expected in view of the energetically low-lying π* orbital of the N–C bond and the consequently strong metal-to-ligand back donation. N–C bond lengths similar to that in **6** have been observed in the few comparable metalla-aza-cyclopropanes.[11] The large change in the ^{13}C NMR spectroscopic resonances ($\Delta\delta = \delta_{complex} - \delta_{ligand}$) of the C=$C_{trop}$ units ($\Delta\delta$ = −75.8 and −67.0 ppm) on coordination to the Rh center in **6** is particularly noteworthy, thus indicating significant cyclopropane character for the RhC_2 units.

Table 1: Selected bond lengths [Å] and angles [°] of **3** (experimental), **6** (experimental), and **6**$^{\bullet+}$ (calculated by DFT).[a]

Compound	Rh-C16	Rh-N	N-C16	Rh-P	Rh-ct	C=C_{trop}	N-Rh-P	ct-Rh-ct	Rh-N-C16	N-C16-Rh	C16-Rh-N
$\mathbf{3}_{exp}$	–	2.131(5)	1.505(7)	2.254(1)	2.075(4)	1.413(6)	158.2(1)	134.5(2)	103.1(3)	–	–
$\mathbf{6}_{exp}$	2.117(4)	2.059(3)	1.446(5)	2.254(1)	2.065(3)	1.419(4)	160.35(9)	132.6(1)	71.9(2)	67.6(2)	40.5(1)
$\mathbf{6}^{\bullet+}_{calcd}$	2.090	2.164	1.417	2.367	2.179	1.422	158.50	136.05	67.75	73.37	38.88

[a] ct indicates the midpoint of the coordinated C=C_{trop} bond.

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A cyclic voltammogram recorded in a THF solution (0.1 M nBu$_4$NPF$_6$ electrolyte, $T = 298$ K, Pt electrode, scan rate 100 mV s^{-1}) of the rhodaazacyclopropane **6** showed one reversible wave at a remarkably low potential ($E^{o}_{1/2} = -0.510$ V vs. the ferrocenium/ferrocene couple), corresponding to the oxidation **6**→**6**$^{\bullet+}$.[12] The high stability of **6**$^{\bullet+}$ allowed the first detailed study of the electronic structure of a rhodaazacyclopropyl radical cation by continuous-wave (CW) and pulse EPR spectroscopy.[13] For this study, **6** as well as the ^{13}C- and D$_2$-labeled derivatives [^{13}C]**6** and [D$_2$]**6** were oxidized with ferrocenium triflate to give the radical cations **6**$^{\bullet+}$, [^{13}C]**6**$^{\bullet+}$, and [D$_2$]**6**$^{\bullet+}$, respectively. The half-life of **6**$^{\bullet+}$ in THF solution (ca. 0.1 mM) is about five hours at room temperature.[14]

The frozen-solution X-band CW-EPR spectra of **6**$^{\bullet+}$ and [^{13}C]**6**$^{\bullet+}$ are shown in Figure 3a. Both spectra are well-resolved at the high-field end ($g_2 = 2.0155$, $g_3 = 2.0145$) and show splittings from the ^{103}Rh nucleus ($I = 1/2$) for **6**$^{\bullet+}$ and from the ^{103}Rh and ^{13}C nuclei ($I = 1/2$) for [^{13}C]**6**$^{\bullet+}$. At the low-field end ($g_1 = 2.121$), the spectra are poorly resolved.[15] Pulse ENDOR (electron nuclear double resonance) and HYSCORE (hyperfine sublevel correlation) spectroscopies were used to determine the complete set of hyperfine parameters of the ^{103}Rh and ^{13}C nuclei of the methylene unit, as well as of the two methylene ^{1}H nuclei, the weakly coupled ^{14}N nucleus, the olefinic ^{13}C nuclei, the olefinic protons, and the ^{31}P nucleus (see the Supporting Information). To assign the methylene proton signals unambiguously, the corresponding spectra of [D$_2$]**6**$^{\bullet+}$ were also recorded (Supporting Information).

In Figure 3b, a representative hyperfine-correlated ENDOR (HYEND) spectrum of [^{13}C]**6**$^{\bullet+}$ is displayed which contains ^{103}Rh and ^{13}C ENDOR frequencies (ν_{ENDOR} axis) that are correlated with their corresponding hyperfine frequencies (ν_{HF} axis). In Figure 3c, a representative Q-band HYSCORE spectrum of **6**$^{\bullet+}$ is shown which contains peaks from the ^{103}Rh nucleus, the ^{13}C nuclei of the olefinic C5 atoms, and the ^{14}N and ^{31}P nuclei. The EPR parameters were obtained from the spectrum by matching the position of the experimental cross-peaks with those calculated from the relevant spin Hamiltonian. A complete set of experimentally derived parameters is given in Table 2, along with the data obtained from DFT calculations.[16] The experimental spin populations were calculated from their respective hyperfine couplings.[17]

Overall, the agreement between experimentally derived and calculated EPR parameters and spin populations is excellent. According to the DFT calculations, the main part of the spin density is located on the rhodium center (47%), the methylene carbon atom (14%), and the four olefinic carbon atoms (C5 2×15%, C4 2×−3%). This distribution of spin density can be seen by inspection of the spin-density plot

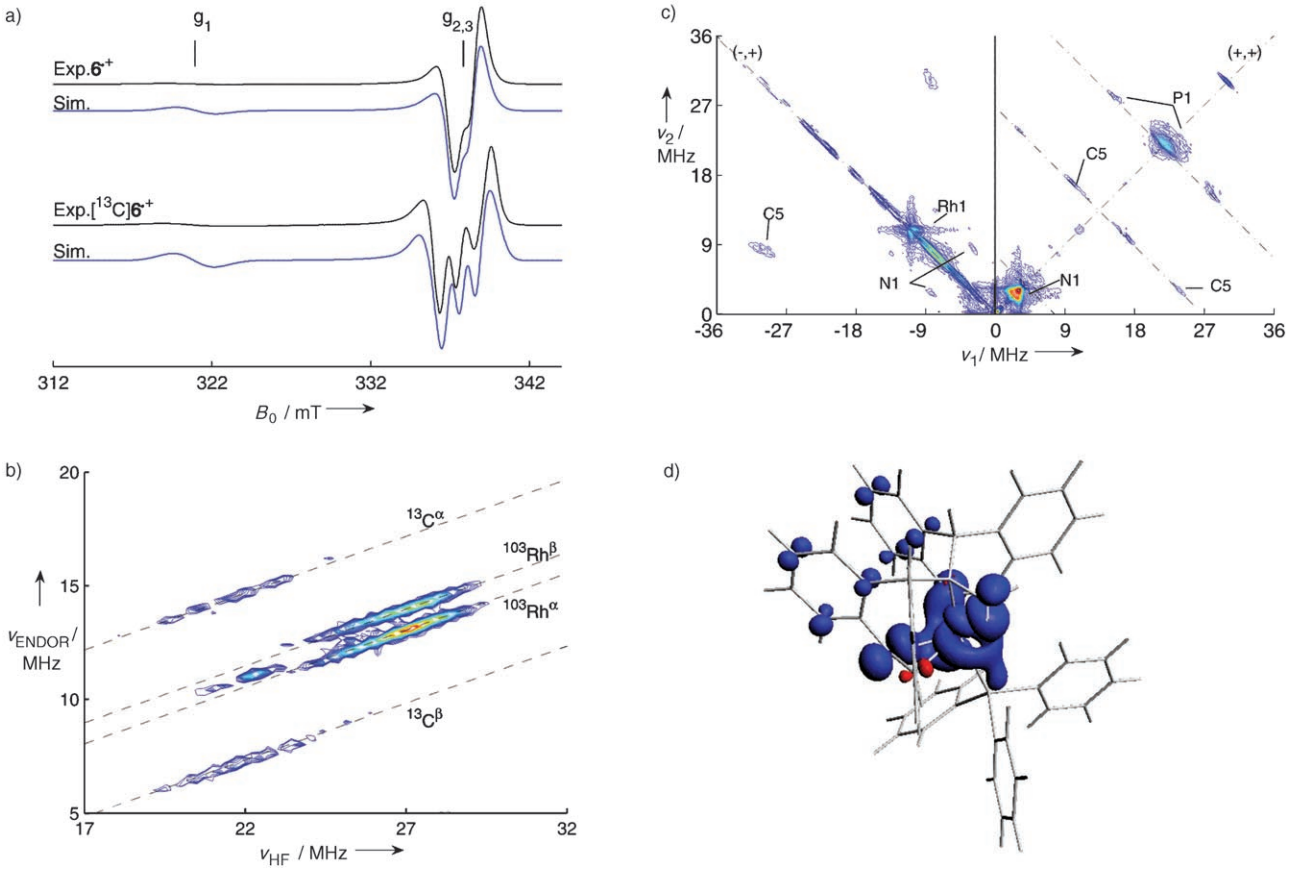


Figure 3. a) Experimental (Exp.) and simulated (Sim.) second-derivative X-band CW-EPR spectra of **6**$^{\bullet+}$ and [^{13}C]**6**$^{\bullet+}$ in THF measured at 115 K. b) X-band HYEND spectrum of [^{13}C]**6**$^{\bullet+}$ in THF measured at 15 K at the observer position g_2. The dashed lines are separated by twice the nuclear Zeeman interaction ($2\nu_I$) of ^{103}Rh and ^{13}C, and cross the ν_{ENDOR} and ν_{HF} axes at ν_I(^{103}Rh) and ν_I(^{13}C), respectively. c) Q-band HYSCORE spectrum of **6**$^{\bullet+}$ in THF measured at 15 K near to the g_3 observer position. Cross-peaks from ^{13}C (in natural abundance), ^{103}Rh, ^{31}P, and ^{14}N are labeled. d) Spin density of compound **6**$^{\bullet+}$, plotted with an isosurface level of 0.004.

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Table 2: EPR parameters and spin populations of compound $6^{\bullet+}$. DFT values are given in brackets.[a]

Group	Nucleus	A_1 [MHz]	A_2 [MHz]	A_3 [MHz]	Spin population
–	Rh1	29 ± 2 (26.3)	25 ± 2 (22.6)	−18 ± 2 (−12.2)	– (47%)
methylene	C16	19 ± 2 (16.3)	20 ± 2 (16.8)	66 ± 2 (47.8)	15% (14%)
methylene	H16	−11 ± 1 (−7.9)	0 ± 2 (0.1)	3 ± 1 (4.2)	2×−0.2% (2×−0.1%)
amine	N1	−1.6 ± 0.2 (−1.6)	−0.9 ± 0.2 (−0.8)	0.2 ± 0.2 (0.4)	−0.5% (−0.5%)
PPh_3	P1	0 to 15 (0.4)	0 to 15 (4.3)	0 to 15 (17)	– (2.5%)
olefin	C5	21 ± 5 (−0.2)	21 ± 5 (−0.1)	55 ± 5 (40.3)	2×11% (2×15%)
	C4	(−8.6)	(−4.2)	(−2.7)	(2×−3%)
olefin	H5	−11 ± 1 (−10.8)	4 ± 2 (1.3)	4 ± 1 (2.3)	2×−0.1% (2×−0.2%)
	H4	5 ± 1 (3.5)	5 ± 2 (5.2)	10 ± 1 (9.3)	2×0.5% (2×0.3%)

[a] The signs of the experimental hyperfine couplings $A_{1,2,3}$ are assigned according to the DFT results; principal axis directions are given in the Supporting Information. $g_1 = 2.121 \pm 0.001$ (2.1050), $g_2 = 2.0155 \pm 0.0005$ (2.0174), $g_3 = 2.0145 \pm 0.0005$ (2.0163).

given in Figure 3d. There is some spin delocalization to the two benzo moieties (2×6%) of the trop ligand, with the alternating sign of the hyperfine couplings of adjacent carbon nuclei indicating a spin-polarization (hyperconjugation) mechanism. Both experimental and calculated data show small hyperfine couplings and thus small spin populations on the nitrogen and phosphorus nuclei. A comparison of **6** and $\mathbf{6}^{\bullet+}$ shows that the gross structural features are retained.

In summary, X-ray, NMR, and EPR spectroscopic data strongly support the proposal that **6** and $\mathbf{6}^{\bullet+}$ are best formulated as RhNC metallacycles. In such a formulation, the interaction of the filled π(C–N) orbital (polarized towards the N atom) with unoccupied metal orbitals represents the Rh–N bond, and the interaction of the unoccupied π*(C–N) orbital (polarized towards the C atom) with occupied metal orbitals represents the Rh–C bond, which lies higher in energy. Therefore, oxidation of **6** mainly affects the Rh–C bond, and consequently the spin population at the carbon atom in $\mathbf{6}^{\bullet+}$ is significant, but very low at the nitrogen atom.

It has been convincingly demonstrated that constraints imposed by the ligand sphere can be used to control the redox potential of transition-metal complexes,[18] and we have reported that four-coordinate d^8 Rh complexes have large negative reduction potentials when the ligand sphere is not flexible enough to allow for a tetrahedral distortion.[19] However, as a result of a lack of suitable reference compounds, the present study does not allow the firm conclusion that small-ring metalla-heterocycles are more easily oxidized than corresponding larger-ring complexes or comparable acyclic compounds, and it remains an open question whether compounds such as **6** are strained molecules. Certainly, the coordination of the $C{=}C_{trop}$ units to the metal center also contributes to the low oxidation potential of **6** and the high stability of its radical cation $\mathbf{6}^{\bullet+}$.[20] An assignment of an oxidation state to the metal center in $\mathbf{6}^{\bullet+}$ is not meaningful,[21] and this complex is best described as a radical with a delocalized unpaired electron.

Experimental Section

Selected spectroscopic data of diagnostic value are given; for details, see the Supporting Information.

3: CH_2Cl_2 (10 mL) and THF (10 mL) were added to [RhCl(trop$_2$NMe)(PPh_3)] (162 mg, 0.200 mmol) and $AgSO_3CF_3$ (51 mg, 0.200 mmol). The resulting red suspension was stirred at room temperature for 1 h and filtered over celite to remove AgCl. The solvents were evaporated in vacuo, and the residue was recrystallized from CH_2Cl_2/hexanes to yield **3** (175 mg, 0.189 mmol, 94%) as red needles. M.p. > 190 °C (decomp); ^{1}H NMR (300.1 MHz, CD_2Cl_2): $\delta =$ 1.85 (d, $^4J_{PH} = 2.9$ Hz, 3H, NCH_3), 4.88 (dd, $^3J_{HH} = 8.9$ Hz, $^3J_{PH} =$ 2.0 Hz, 2H, CH_{olefin}), 6.61 ppm (ddd, $^3J_{HH} = 8.9$ Hz, $^3J_{PH} = 2.9$ Hz, $^2J_{RhH} = 2.9$ Hz, 2H, CH_{olefin}); ^{13}C NMR (75.5 MHz, CD_2Cl_2): $\delta = 44.6$ (s, NCH_3), 78.6 (d, $^1J_{RhC} = 7.3$ Hz, CH_{olefin}), 88.5 ppm (d, $^1J_{RhC} =$ 13.1 Hz, CH_{olefin}); ^{31}P NMR (121.5 MHz, CD_2Cl_2): $\delta = 39.2$ ppm (d, $^1J_{RhP} = 138$ Hz); ^{103}Rh NMR (12.6 MHz, $CDCl_3$): $\delta = 1272$ ppm (d, $^1J_{RhP} = 138$ Hz); HRMS (MALDI): *m/z*: 776.1975 (calcd for $C_{49}H_{40}NPRh^+$: 776.1948).

6: A solution of KO*t*Bu (18.2 mg, 0.162 mmol) in THF (2 mL) was added to a suspension of **3** (150 mg, 0.162 mmol) in THF (5 mL). The resulting orange solution was filtered over a plug of alumina and concentrated to 1 mL under reduced pressure. The product crystallized upon addition of toluene (1 mL) and hexanes (10 mL). The mother liquor was decanted, and orange crystals of **6** were dried in vacuo. Yield: 88 mg (0.114 mmol, 70%); m.p. > 165 °C (decomp); ^{1}H NMR (400.1 MHz, [D$_8$]THF): $\delta = 2.89$ (dd, $^3J_{PH} = 3.3$ Hz, $^2J_{RhH} =$ 2.0 Hz, 2H, CH_2), 3.83 (ddd, $^3J_{HH} = 9.5$ Hz, $^3J_{PH} = 8.5$ Hz, $^2J_{RhH} =$ 1.5 Hz, 2H, CH_{olefin}), 4.37 ppm (dd, $^3J_{HH} = 9.5$ Hz, $^2J_{RhH} = 2.0$ Hz, 2H, CH_{olefin}); ^{13}C NMR (62.9 MHz, [D$_8$]THF): $\delta = 54.7$ (d, $^1J_{RhC} =$ 9.7 Hz, CH_{olefin}), 62.4 (d, $^1J_{RhC} = 17.0$ Hz, CH_2), 63.5 ppm (d, $^1J_{RhC} =$ 8.7 Hz, CH_{olefin}); ^{31}P NMR (162.0 MHz, [D$_8$]THF): $\delta = 63.8$ ppm (d, $^1J_{RhP} = 185$ Hz); ^{103}Rh NMR (12.6 MHz, [D$_8$]THF): $\delta = -229$ ppm (d, $^1J_{RhP} = 185$ Hz); HRMS (MALDI): *m/z*: 776.1937 (calcd for $C_{49}H_{40}NPRh^+$: 776.1948); EPR samples of $\mathbf{6}^{\bullet+}$ were prepared by adding a solution containing one equivalent of ferrocenium triflate (5 mM in THF) to a solution of **6** (5 mM in THF). Sample tubes were degassed by three freeze-pump-thaw cycles, sealed, and stored in liquid nitrogen.

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[6] **3**·3 CH_2Cl_2: $C_{53}H_{44}Cl_6F_3NO_3PRhS$, red needles, orthorhombic, space group *Pnma*, $a = 23.791(4)$, $b = 17.642(3)$, $c = 12.192(2)$ Å, $V = 5117(1)$ Å^3, $Z = 4$, $\rho_{calcd} = 1.530$ Mg m^{-3}, $Mo_{K\alpha}$, 200 K, $2\theta_{max} = 52.74°$, 30414 reflections, 5403 independent ($R_{int} = 0.0679$), Patterson, empirical absorption correction SADABS (version 2.03), refinement against full matrix (versus F^2) with SHELXTL (version 6.12) and SHELXL-97, 341 parameters, $R1 = 0.0598$, $wR2 = 0.1568$ (all data), max./min. residual electron density 1.339/−1.624 e Å^{-3}. **6**·toluene: $C_{56}H_{47}NPRh$, orange crystals, orthorhombic, space group *Pnma*, $a = 25.397(1)$, $b = 13.634(1)$, $c = 12.052(1)$ Å, $V = 4173.0(4)$ Å^3, $Z = 4$, $\rho_{calcd} = 1.381$ Mg m^{-3}, $Mo_{K\alpha}$, 250 K , $2\theta_{max} = 52.74°$, 25675 reflections, 4450 independent ($R_{int} = 0.0711$), direct methods, refinement against full matrix (versus F^2) with SHELXTL (version 6.12) and SHELXL-97, 291 parameters, $R1 = 0.0415$, $wR2 = 0.0933$ (all data), max./min. residual electron density 0.823/−0.594 e Å^{-3}. CCDC-290105 (**3**) and CCDC-290104 (**6**) contain the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif. For further details, see also the Supporting Information.

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[15] The poor resolution is probably a result of considerable g strain caused by a distribution of different molecular conformations and poor glass formation in frozen solution.

[16] DFT calculations were performed with the Amsterdam Density Functional (ADF 2005.01) package. Details are given in the Supporting Information.

[17] J. R. Morton, K. F. Preston, *J. Magn. Reson.* **1978**, *30*, 577.

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